



Introduction to Computational Social Science

Institute of Sociology and Demography (ISD),
University of Rostock,
Germany

Instructor Info —



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Office Hrs: by email agreement



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Course Info —



Pre-requirement: None



Thursdays



11:15-12:45



PC pool 227, ISD

Lab Info —



Every other week



11:15-12:45



PC pool 227, ISD

Overview

Computational Social Science (CSS) is a scientific discipline where computational tools and techniques are used to answer research questions that could be social in nature. This area has attracted two groups of scientists: a) social scientists with computational skills, and b) computational scientists with an interest in questions related to social phenomena. The advent of social media and online social networks has led to increased digitization of human interactions and exponential growth of the digital traces left behind in these interactions. Both of these groups of scientists know the value of these digital trace data and have the skills to analyze them.

Generally, research projects, including in CSS, could follow two main approaches: data-driven (inductive) or theory-driven (deductive). While the data-driven approach might start from an inductive or bottom-up exploration of the data to find general patterns, the theory-driven approach –predominantly used in social sciences– usually starts with a question and follows a few steps: Question and theorize, Gather and pre-process data, Model, and Report and publish.

Learning Objectives

By the end of this course, students will be familiar with deductive and inductive approaches to research. They will have knowledge of different research questions, data sources, and possible answers for the questions provided in Computational Social Science. Through hands-on lab sessions, they will acquire or advance their technical skills in R, Python, and SQL to use large-scale digital trace data. They will also learn about ethics in research using digital trace data.

Reading Materials

Every week's materials include required and recommended readings specific to that week's topic, and some of the main ones are listed below.

Required Text

Salganik, M. J. (2018). *Bit by bit: Social research in the digital age*. Princeton University Press.

Recommended Texts

Akbaritabar, A. (2024). Thinking spatially in computational social science. *EPJ Data Science*, 13(1), 1–12. <https://doi.org/10.1140/epjds/s13688-023-00443-0>

Cioffi-Revilla, C. (2017). *Introduction to Computational Social Science*. Springer International Publishing. <https://doi.org/10.1007/978-3-319-50131-4>

Fortunato, S., Bergstrom, C. T., Börner, K., Evans, J. A., Helbing, D., Milojević, S., Petersen, A. M., Radicchi, F., Sinatra, R., Uzzi, B., Vespignani, A., Waltman, L., Wang, D., & Barabási, A.-L. (2018). Science of science. *Science*, 359(6379), eaao0185. <https://doi.org/10.1126/science.aao0185>

Kashyap, R., Rinderknecht, R. G., Akbaritabar, A., Alburez-Gutierrez, D., Gil-Clavel, S., Grow, A., Kim, J., Leasure, D. R., Lohmann, S., Negraia, D. V., Perrotta, D., Rampazzo, F., Tsai, C.-J., Verhagen, M. D., Zagheni, E., & Zhao, X. (2022). *Digital and Computational Demography* (tech. rep.). SocArXiv. <https://doi.org/10.31235/osf.io/7bvpt>

Lazer, D., Pentland, A., Adamic, L., Aral, S., Barabasi, A.-L., Brewer, D., Christakis, N., Contractor, N., Fowler, J., Gutmann, M., Jebara, T., King, G., Macy, M., Roy, D., & Van Alstyne, M. (2009). Computational Social Science. *Science*, 323(5915), 721–723. <https://doi.org/10.1126/science.1167742>

Lazer, D. M. J., Pentland, A., Watts, D. J., Aral, S., Athey, S., Contractor, N., Freelon, D., Gonzalez-Bailon, S., King, G., Margetts, H., Nelson, A., Salganik, M. J., Strohmaier, M., Vespignani, A., & Wagner, C. (2020). Computational social science: Obstacles and opportunities. *Science*, 369(6507), 1060–1062. <https://doi.org/10.1126/science.aaz8170>

Macy, M. W., & Willer, R. (2002). From Factors to Actors: Computational Sociology and Agent-Based Modeling. *Annual Review of Sociology*, 28(1), 143–166. <https://doi.org/10.1146/annurev.soc.28.110601.141117>

FAQs

? How to access the course materials?

! There is a private GitHub repository where all students are added in the first session. Every week's slides, hands-on materials, code, and data will be uploaded there.

? What to do at home before/after sessions?

! 1) Study the reading materials and prepare some questions or discussion points. 2) Run codes in hands-on materials, reproduce the examples, extend them, and note the unclear points.

? Do I need programming skills to attend? Where to learn the basics of R, Python, SQL?

! No background knowledge is required, but some familiarity will help you advance faster. The course materials cover the basics and more advanced topics. Start by practicing with the provided examples. Materials also provide links to other introductory courses.

? Can we use Artificial Intelligence to help us?

! AI is a tool similar to any other tool out there. You should be "in control", and take "responsibility" for using it. Follow the ISD institute's reporting protocol to share how you used it.

? Where to learn about academic writing?

! 1) <https://writingcenter.gmu.edu/writing-resources/imrad/writing-an-imrad-report>
2) <https://www.nature.com/scitable/topicpage/effective-writing-13815989/>

Grading Scheme

The final grade will be on a 1-6 scale with these meanings: 1.0: "sehr gut"/ Excellent; 1.3: "sehr gut"/ Excellent; 1.7: "gut"/ Good; 2.0: "gut"/ Good; 2.3: "gut"/ Good; 2.7: "befriedigend"/ Satisfactory; 3.0: "befriedigend"/ Satisfactory; 3.3: "befriedigend"/ Satisfactory; 3.7: "ausreichend"/ Pass; 4.0: "ausreichend"/ Pass.

Active participation in course discussions during the semester will inform my evaluation of your understanding of the topics. If you have specific concerns that might inhibit your active participation in class discussions of the reading materials, please inform me at the start of the course.

Course Examination

The mode of examination is a "term paper" ("Hausarbeit" in German). In the penultimate week, I will share a list of topics (from the course schedule below) and ask you to select one. You will use the reading list for that topic, search for other literature, plus use the data sources and discussed research questions, and write an essay that should include these sections: a) introduction, b) background and literature, c) methods and analytical strategy, d) results, e) discussion and conclusion (including possible limitations of your method and results), f) references (to the literature, data sources, etc., that you used).

Diversity and Inclusivity Statement

To become a great social scientist, a student needs to learn to add a "why" before every statement. The same statements that others hear without questioning. Students need to develop sharp observation skills and find a new angle in seeing familiar social phenomena. This is how they can observe inequalities and previously less explored aspects of society. This is the same society in which we all live, however, those with sharpened observation skills can identify inequalities in a familiar context. They can question the ordinary, find reasons behind observed phenomena, and imagine counterfactual scenarios for the future. This practice helps students develop critical thinking skills. I converse with students, write their ideas on the whiteboard, and ask them to explain those ideas. I create a safe space where every student feels welcome to contribute, which helps students develop their own ideas confidently.

Students with Special Needs

Please inform me as soon as possible of special needs that you may have. The sooner you notify me, the better I will be able to coordinate with the institute and other responsible persons to accommodate your special needs.

Students' Mental Health and Well-being

Academic life and university studies can be very stressful and challenging. Please inform me or seek proper professional help if you are concerned with your mental health and well-being. Seeking help is one of the most courageous things to do.

Integrity, Plagiarism, and Ethics of Research

Students are expected to prepare their final assignments independently. Any violation of this will not be tolerated and will be appropriately reported to the faculty and university according to the institution's guidelines. If you use a resource in your work, cite it properly and give due credit by following established guidelines, e.g., APA: <https://apastyle.apa.org/style-grammar-guidelines/citations/plagiarism>.

Credits

- The logo of the course is designed using Google Gemini.
- The syllabus $\LaTeX$ template is modified based on <https://www.overleaf.com/latex/templates/inzane-syllabus-template/zpgzsdxfvqmz>.

Class and Lab Schedule

PART 1: Introductory topics

Week 1	Introduction	What is "Computational Social Science"? Inductive and deductive research; Digitalization; Big data revolution; Designed (thick) data tailored for research, such as survey and census; Register data
Week 2	Open Science and Reproducibility	Reproducibility crisis; Pre-registration; Version control, Git, GitHub, Accessing course materials
Week 3	Digital Trace Data	Observational data; Available vs. designed data; APIs and web scrapping; Representativeness

PART 2-A: Empirical topics and cases

Week 4	Mobility and Migration	Computational approaches to migration research; Spatial representation of the social world; The Scholarly Migration Database
Week 5	Network Analysis	Migration networks; Relational representation of the social world; Tie formation mechanisms in social networks; Violence of independence of observations
Week 6	Science of Science	Robert K. Merton; Sociology of scientific knowledge vs Bibliometrics/ Scientometrics
Week 7	Text as Data	Natural Language Processing; Topic modelling; LDA; Structural Topic Models; Word embedding vectors; Spatial view of text data

PART 2-B: (Conditional on time)

Week 8	Social Simulation	Agent-based modelling for social scientists; Micro-Macro link and Coleman's boat; Empirical calibration of simulation models; Differences of ABMs with Microsimulation; ODD protocol
Week 9	Machine Learning	Supervised and unsupervised use of observational data; Feature learning

PART 3: Conclusions

Week 10	"Other" CSS Skills	Parallelization; Functional vs Object-Oriented; Graph databases; DuckDB; SnakeMake Workflows
Week 11	Ethics in Computational Social Science	Informed consent; Personal data; GDPR; Biases in Science
Week 12	Limitations of Computational Social Science	Pitfalls of digital trace data and computational approaches; Representativeness; AI and the future of Computational Social Science
	For final examination	LIST OF FINAL ASSIGNMENTS
Week 13	Conclusions	Concluding remarks and discussion; Thick vs Big data, Survey experiments, and mixed methods; Linked data
	Discussing final examination	Share your selected topic; 6 weeks to prepare a short essay and empirical analysis on one of the CSS topics/skills